Grade 10 · Algebra II · Topic 2: Quadratic Functions and Equations · Lesson 2-5 — Completing the Square

Project Task — Corniche Water Jets

A 20-minute group project. Work in fours. Every member signs the sheet.

Student name :	Date : .....
Group members :	

MATERIALS This sheet · Squared paper · A pencil and ruler · A calculator

THE DRIVING QUESTION

Two decorative water jets on the Jeddah Corniche follow different paths. Which jet reaches the greater height — and which one stays in the air longer?

THE SITUATION

A classroom model (invented for practice, not a design): Jet 1 has height $h = -5t^2 + 20t + 1$ metres and Jet 2 has height $h = -4t^2 + 24t + 3$ metres, where t is the time in seconds after leaving the nozzle.

Designers need, for each jet, the greatest height, the time it happens and the time it reaches sea level ($h = 0$). Each group recommends the jet for a high display and the jet for a long display.

$$h_1 = -5t^2 + 20t + 1 \qquad h_2 = -4t^2 + 24t + 3$$

Jet	Greatest height (m)	At time (s)	Sea level at (s)
1			
2			

YOUR TASK 5 steps · 20 minutes

1 — COMPLETE THE SQUARE (5 min)
Take out the leading coefficient and write each height in the form $a(t - p)^2 + q$.

2 — READ THE VERTEX (3 min)
State the greatest height and the time it happens for each jet. Fill the table.

3 — REACH THE SEA (5 min)
Set $h = 0$, solve by completing the square and reject the negative root. Round to the nearest hundredth.

4 — COMPARE (4 min)
Which jet is higher? Which stays in the air longer? Check each by substituting your time into the model.

5 — PRESENT (3 min)
One sentence for the designer: why does the vertex form show the greatest height at a glance?

YOUR WORKING write on this sheet

Step 1 — the vertex forms:

Step 2 — the greatest heights and times:

Step 3 — the times at sea level:

Steps 4 and 5 — our comparison and sentence:

ROLES

- Reader — states the models and the question
- Calculator — completes the squares
- Checker — substitutes answers into the model
- Presenter — says the sentence

DONE WHEN ...

- The adjustment term is balanced outside the bracket.
- The negative root is rejected with a reason.
- Units (metres, seconds) appear in every answer.
- Your sentence makes sense to a non-mathematician.